Power Supply Systems

Part II

Phase 10: Earthing, Wiring and Electrical Installation

Aim: To complete the remaining Part-II syllabus: earthing, service mains, wiring, installation testing and special lighting connections.

Style: Simple English, SN ma'am style, clean diagrams, no overlapping labels, every diagram word explained.

Target: Pipe earthing, plate earthing, earth resistance, wiring systems, domestic installation, service mains, testing and special lighting diagrams.

Subject:	Power Supply Systems
Section:	Part II
Phase:	10 – Earthing, Wiring and Installation
Level:	Absolute zero to semester-question-ready

Academic article style notes

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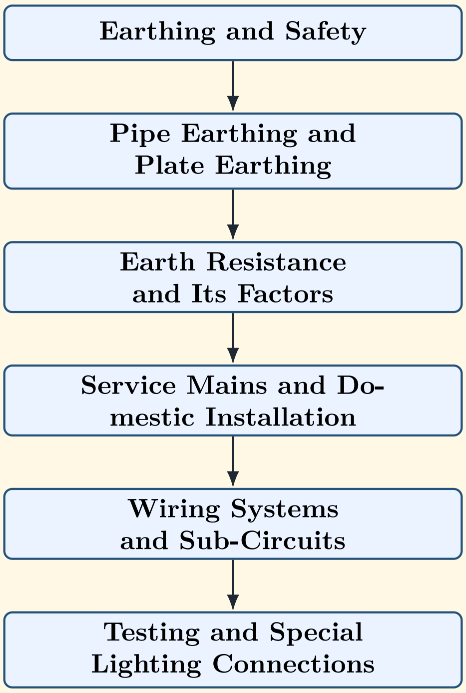
1 Phase 10 Roadmap

Phase 10 Goal

This phase completes the practical side of Part II. It covers how an electrical installation is made safe, wired, connected to supply, tested and operated.

Phase 10 Learning Flow



How to Read the Diagram

Word	Meaning
Earthing and Safety	Connecting metal parts to earth so that fault current gets a safe path.
Pipe Earthing and Plate Earthing	Two common practical earthing methods.
Earth Resistance	Resistance offered by earth electrode and surrounding soil to fault current.
Service Mains	Conductors connecting supplier's line to consumer's installation.
Wiring Systems	Methods of running wires inside buildings.
Testing and Special Lighting	Safety tests and common lamp-control circuits.

2 Earthing

Definition

Earthing means connecting the non-current-carrying metallic parts of electrical equipment to the general mass of earth through a low-resistance conductor.

Earthing provides a safe path for fault current.

Earthing protects human life and equipment.

2.1 Need of Earthing

Earthing is required to:

1. protect people from electric shock,
2. protect equipment from damage,
3. provide low-resistance path for fault current,
4. operate protective devices quickly,
5. keep metal bodies at earth potential,
6. protect installation from leakage current,
7. reduce fire risk.

Memory Hook

Earthing idea:

Fault current should go to earth, not through a human body.

3 Equipment Earthing and System Earthing

3.1 Equipment Earthing

Definition

Equipment earthing means connecting the metallic body of equipment to earth.

Example:

motor frame, transformer tank, switchboard body, appliance body

3.2 System Earthing

Definition

System earthing means connecting one point of the electrical system, usually neutral, to earth.

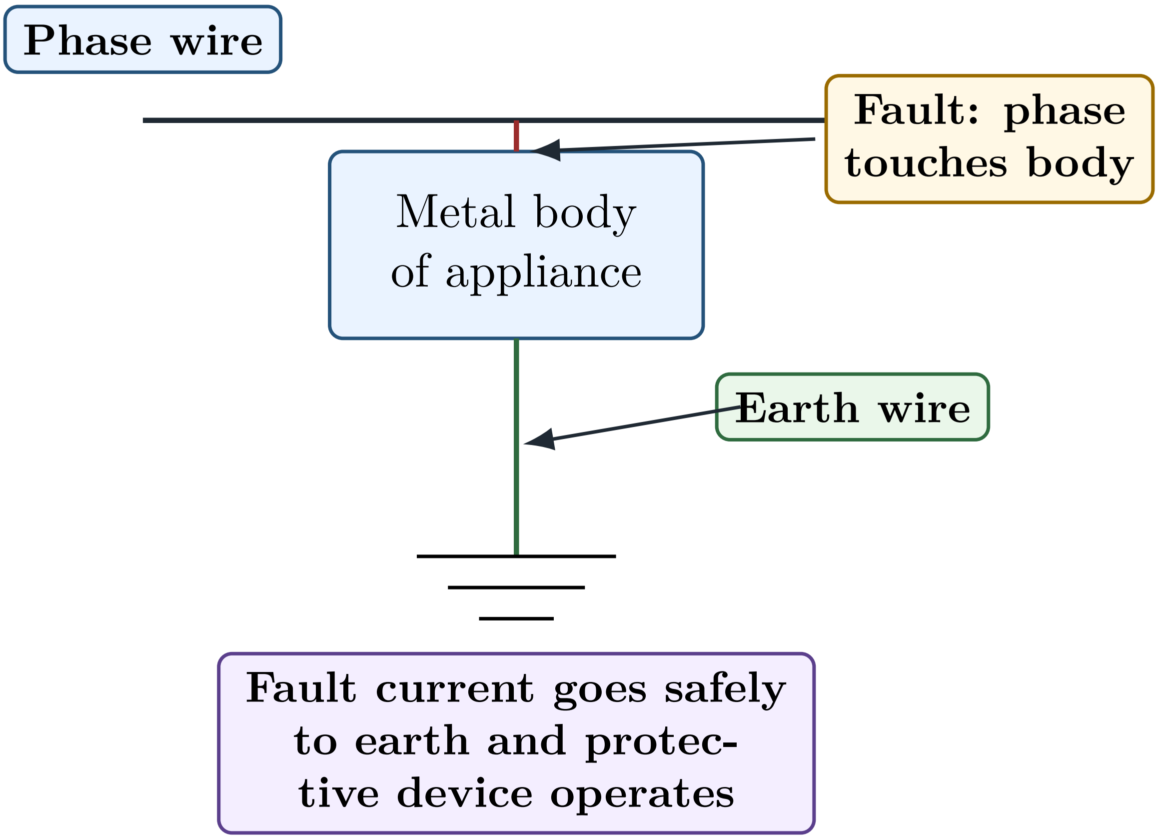
Example:

neutral of transformer or generator connected to earth

Point	Equipment Earthing	System Earthing
What is earthed?	Metallic body of equipment	Neutral or one point of system
Purpose	Human safety	Voltage stability and fault protection

Example	Motor body earthed	Transformer neutral earthed
Current in normal condition	No current normally	May carry unbalanced or fault current depending system

4 Basic Earthing Safety Diagram



How to Read the Diagram

Word/Symbol	Meaning
Phase wire	Live conductor supplying the appliance.
Metal body of appliance	Non-current-carrying metallic part that may become live during fault.
Fault: phase touches body	Insulation failure causing live wire to touch metal body.
Earth wire	Low-resistance wire connecting metal body to earth.
Earth symbol	Represents connection to the general mass of earth.
Fault current path	Current flows to earth through earth wire, so shock risk is reduced.

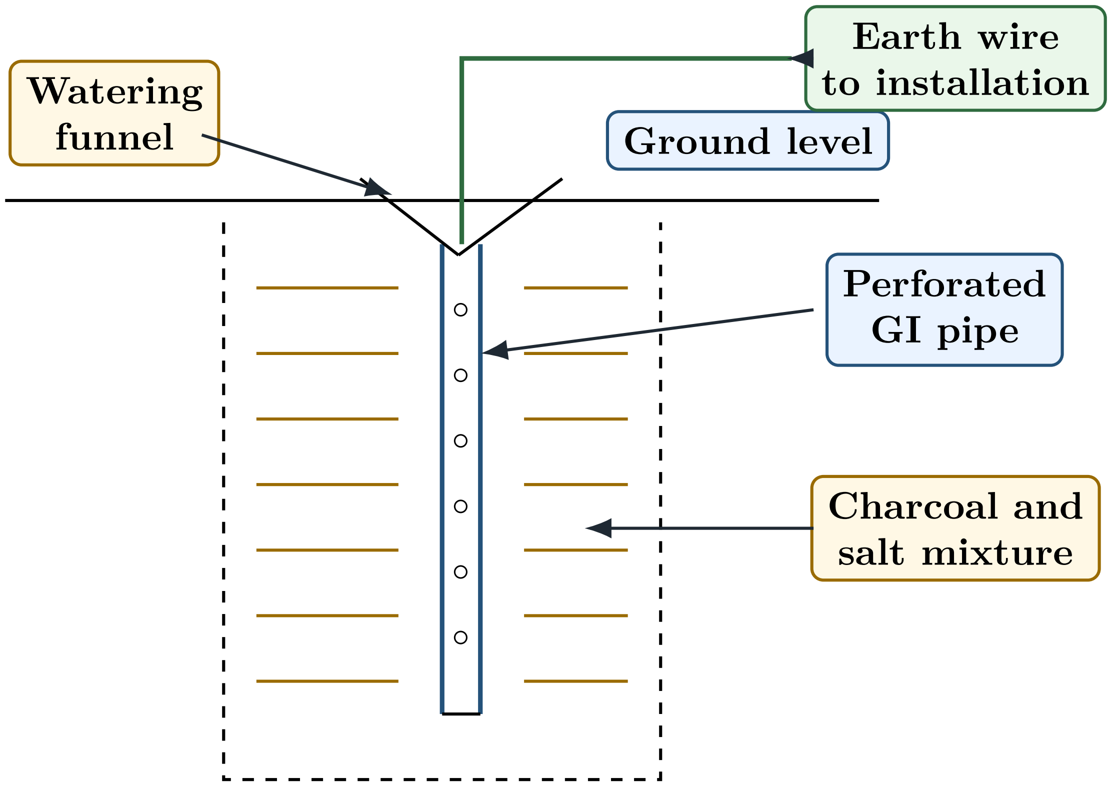
5 Pipe Earthing

Definition

Pipe earthing is a method of earthing in which a perforated galvanised iron pipe is buried vertically in the ground and connected to the installation by an earth wire.

Pipe earthing is widely used because it is economical and gives good contact with moist soil.

6 Diagram of Pipe Earthing



Pipe Earthing

How to Read the Diagram

Word/Symbol	Meaning
Ground level	Surface of the earth. The earthing pipe is buried below this level.
Perforated GI pipe	Galvanised iron pipe with holes. It acts as the earth electrode.
Holes in pipe	Allow moisture to enter and improve earth contact.
Charcoal and salt mixture	Improves conductivity of soil around the pipe and reduces earth resistance.
Watering funnel	Used to pour water so that soil remains moist.
Earth wire to installation	Conductor connecting electrical installation to the buried pipe.
Pipe Earthing	Complete arrangement of pipe electrode, soil treatment and earth connection.

6.1 Advantages of Pipe Earthing

1. Economical.
2. Easy to install.
3. Requires less space.
4. Gives good contact with moist soil.
5. Commonly used for domestic and commercial installations.

7 Plate Earthing

Definition

Plate earthing is a method of earthing in which a copper or galvanised iron plate is buried vertically in the ground and connected to the installation by an earth wire.

The plate is surrounded by layers of charcoal and salt.

8 Diagram of Plate Earthing

How to Read the Diagram

Word/Symbol	Meaning
Ground level	Surface of the earth. The plate is buried below this level.
Earth Plate	Copper or GI plate acting as earth electrode.
Copper or GI plate	Conducting plate used to make contact with earth.
Charcoal and salt layers	Surrounding material used to reduce earth resistance.
Watering pipe	Pipe used to add water and maintain soil moisture.
Earth wire to installation	Conductor joining the earth plate to the electrical installation.
Plate Earthing	Complete earthing arrangement using buried plate electrode.

9 Pipe Earthing vs Plate Earthing

Point	Pipe Earthing	Plate Earthing
Electrode used	Perforated GI pipe	Copper or GI plate
Cost	Cheaper	Costlier
Space required	Less	More
Installation	Easier	More labour required
Common use	Domestic and general installations	Large installations and important equipment
Maintenance	Watering may be required	Watering may be required

10 Earth Resistance

Definition

Earth resistance is the resistance offered by the earth electrode and surrounding soil to the flow of current into the general mass of earth.

Low earth resistance is desirable because fault current should pass easily to earth.

$$\boxed{\text{Lower earth resistance} \Rightarrow \text{better earthing}}$$

10.1 Factors Affecting Earth Resistance

Earth resistance depends on:

1. soil resistivity,
2. moisture content of soil,
3. temperature,
4. depth of electrode,
5. size of electrode,

6. number of electrodes,
7. distance between electrodes,
8. presence of salt and charcoal,
9. quality of earth connection.

10.2 Methods to Reduce Earth Resistance

1. Increase depth of electrode.
2. Use larger electrode.
3. Use multiple electrodes in parallel.
4. Maintain moisture by watering.
5. Use charcoal and salt around electrode.
6. Improve soil treatment.
7. Use proper joints and clamps.

Semester Answer Style

For a 5-mark earthing answer:

$$\boxed{\text{Definition} \rightarrow \text{Need} \rightarrow \text{Diagram} \rightarrow \text{Factors} \rightarrow \text{Methods to reduce resistance}}$$

11 Service Mains

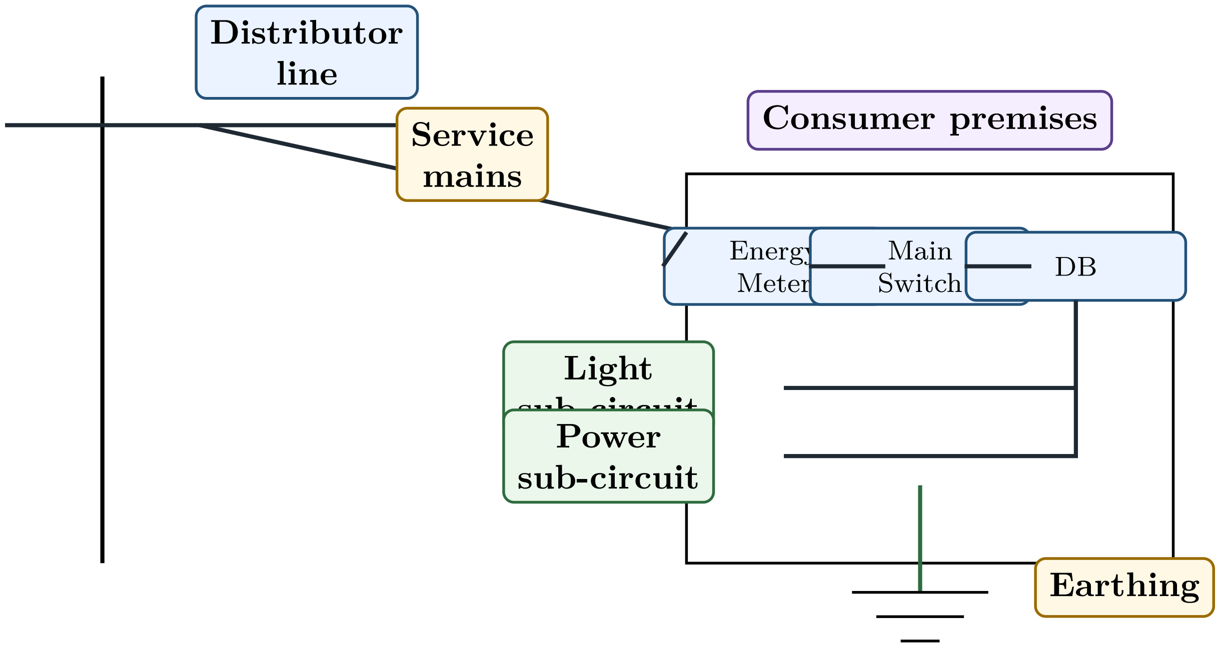
Definition

Service mains are the conductors used to connect the supplier's distribution line to the consumer's premises.

Service mains may be:

- overhead service mains,
- underground service mains.

12 Diagram of Domestic Service Connection



Domestic Service Connection

How to Read the Diagram

Word/Symbol	Meaning
Distributor line	Supplier's low-voltage distribution line.
Service mains	Conductors from distributor line to consumer premises.
Consumer premises	Building where electrical supply is used.
Energy Meter	Measures electrical energy consumed by the consumer.
Main Switch	Main control switch for the whole installation.
DB	Distribution board. It distributes supply to sub-circuits.
Light sub-circuit	Circuit supplying lamps and fans.
Power sub-circuit	Circuit supplying power sockets and heavier loads.
Earthing	Safety connection from installation metal parts to earth.

13 Wiring System

Definition

A **wiring system** is the method of installing electrical conductors and accessories inside a building to supply electrical loads safely.

A good wiring system should be:

1. safe,
2. reliable,

3. economical,
4. durable,
5. easy to inspect,
6. neat in appearance,
7. mechanically protected,
8. easy to extend and repair.

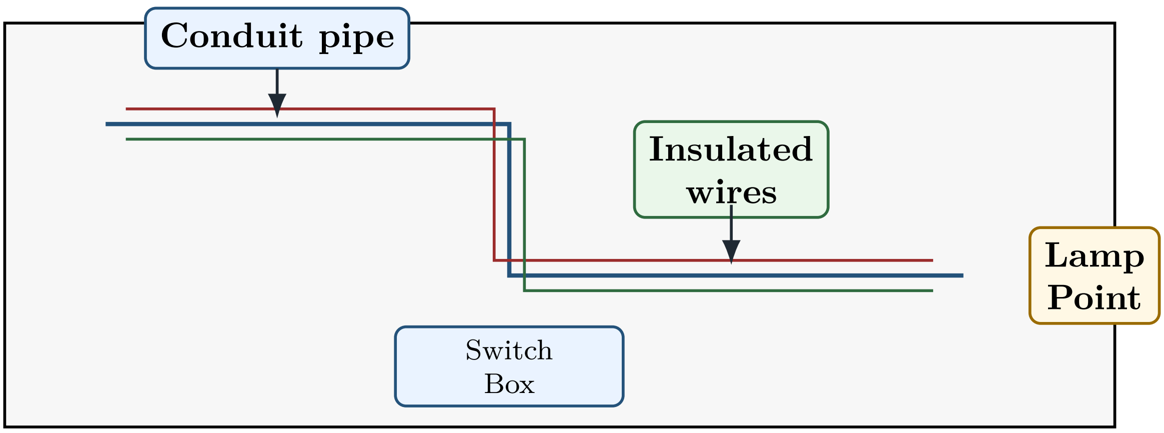
14 Types of Wiring Systems

Important wiring systems are:

1. Cleat wiring,
2. Batten wiring,
3. Casing and capping wiring,
4. Conduit wiring,
5. Concealed conduit wiring.

Wiring Type	**Main Feature**	**Use**
Cleat wiring	Wires are supported on porcelain cleats	Temporary wiring
Batten wiring	Wires are fixed on wooden batten using clips	Small installations
Casing and capping	Wires run inside wooden or PVC casing and covered by capping	Old domestic wiring
Conduit wiring	Wires run inside metal or PVC conduit pipes	Permanent and safe installation
Concealed conduit	Conduits are hidden inside walls	Modern buildings

15 Conduit Wiring Diagram



Conduit Wiring

How to Read the Diagram

Word/Symbol	Meaning
Wall	Building wall where wiring is installed.
Conduit pipe	Protective pipe through which wires pass.
Insulated wires	Phase and neutral wires running inside conduit.
Switch Box	Box where switch is mounted.
Lamp Point	Point where lamp or light fitting is connected.
Conduit Wiring	Safe wiring system in which wires are mechanically protected.

16 Domestic Electrical Installation

A domestic installation generally contains:

1. service mains,
2. energy meter,
3. main switch,
4. fuse or MCB,
5. distribution board,
6. light and fan sub-circuits,
7. power sub-circuits,
8. earthing,
9. switches and sockets.

16.1 Sub-Circuit Concept

Definition

A **sub-circuit** is a branch circuit taken from the distribution board to supply a group of loads.

Common sub-circuits are:

Lighting sub-circuit

Power sub-circuit

16.2 Light and Power Sub-Circuit Difference

Point	Lighting Sub-Circuit	Power Sub-Circuit
Loads	Lamps, fans, small loads	Power sockets, heaters, motors
Current rating	Lower	Higher
Wire size	Smaller	Larger
Protection	Smaller fuse or MCB	Higher-rated fuse or MCB

Use	General lighting and fan points	Heavy appliances

17 Basic Estimation of Wiring Points

For exam estimation-type questions, use this basic sequence:

1. Count number of light points.
2. Count number of fan points.
3. Count number of socket outlets.
4. Separate light sub-circuits and power sub-circuits.
5. Select wire size according to current rating.
6. Select switches, sockets, MCBs and distribution board.
7. Add earthing conductor.
8. Prepare material list.

Semester Answer Style

For a small estimation answer, write:

$$\boxed{\text{Load points} \rightarrow \text{Sub-circuits} \rightarrow \text{Wire size} \rightarrow \text{Protective device} \rightarrow \text{Material list}}$$

18 Testing of Electrical Installation

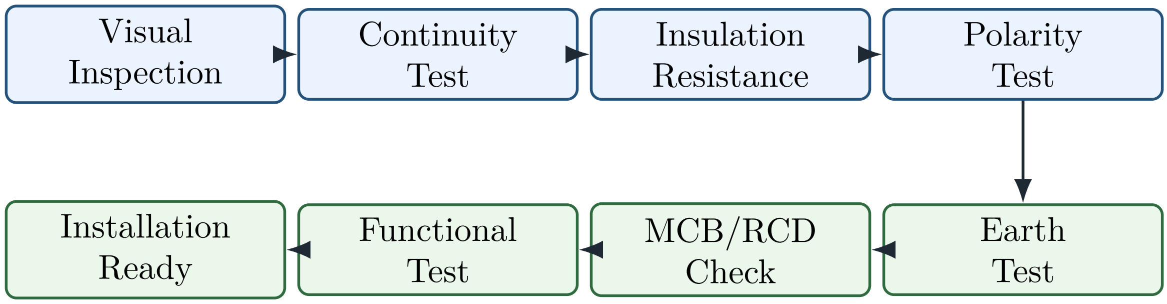
Definition

Testing of installation means checking an electrical installation before energising it to ensure safety, continuity, insulation and correct connections.

Important tests are:

1. Continuity test of protective conductor,
2. Insulation resistance test,
3. Polarity test,
4. Earth continuity test,
5. Earth electrode resistance test,
6. Leakage current test,
7. Functional test of switches and protective devices.

19 Installation Testing Flowchart



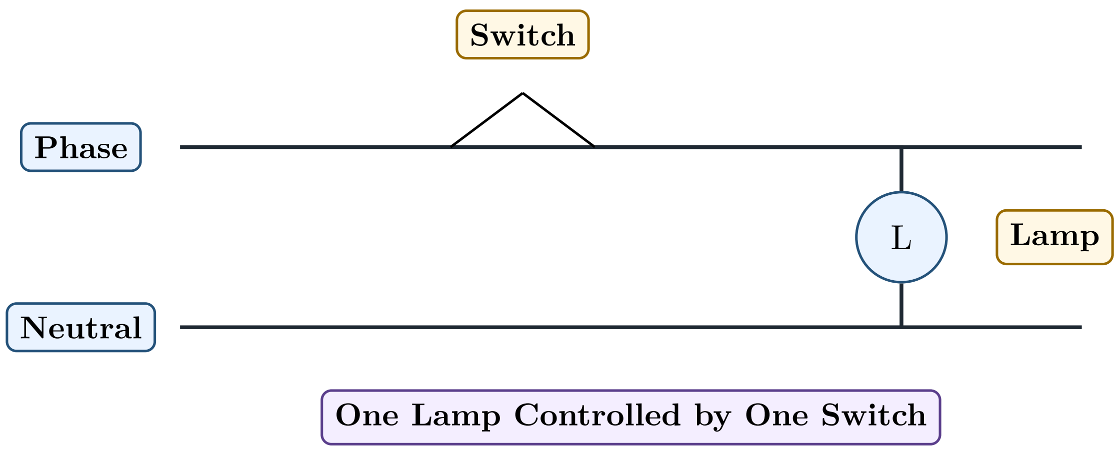
How to Read the Diagram

Step	Meaning
Visual Inspection	Checking visible wiring, joints, switchgear, colour coding and damage.
Continuity Test	Checking that conductors are not broken.
Insulation Resistance	Checking that insulation is healthy and leakage is low.
Polarity Test	Checking that switch and fuse are connected in phase wire, not neutral wire.
Earth Test	Checking earth continuity and earth electrode resistance.
MCB/RCD Check	Checking protective devices operate correctly.
Functional Test	Checking switches, lamps, sockets and appliances operate properly.
Installation Ready	Installation can be energised safely after tests are satisfactory.

20 Special Lighting Connection 1: One Lamp Controlled by One Switch

Definition

A **one-way lighting connection** is a circuit in which one lamp is controlled by one single-pole switch.



How to Read the Diagram

Word/Symbol	Meaning
Phase	Live wire. Switch must be connected in this wire.
Neutral	Return wire connected directly to lamp.
Switch	Opens or closes phase wire to control lamp.
Lamp	Lighting load. It glows when circuit is complete.
One Lamp Controlled by One Switch	Basic lighting circuit used in rooms.

21 Special Lighting Connection 2: Staircase Wiring

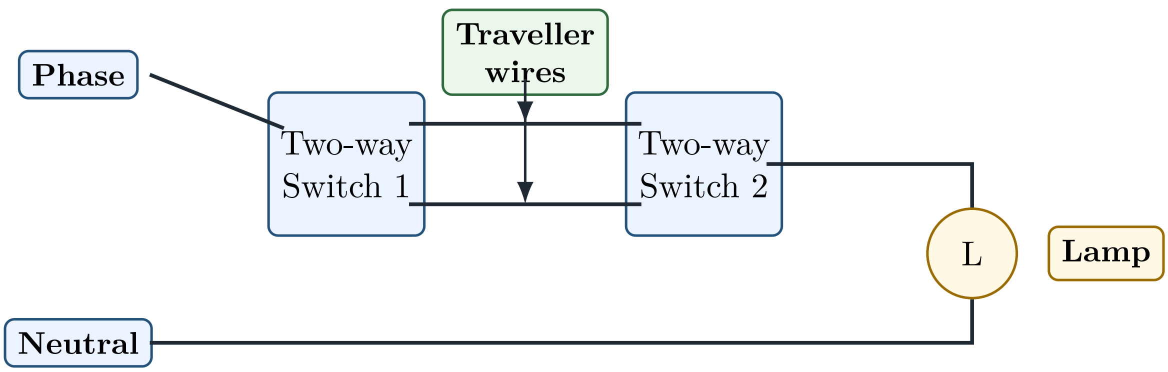
Definition

Staircase wiring is a lighting circuit in which one lamp can be controlled from two different places using two two-way switches.

It is used in:

staircases, corridors, long passages, bedrooms

22 Staircase Wiring Diagram



Staircase Wiring: One Lamp Controlled from Two Places

How to Read the Diagram

Word/Symbol	Meaning
Phase	Live wire entering the first two-way switch.
Neutral	Return wire connected to the lamp.
Two-way Switch 1	First switch, placed at one location such as bottom of staircase.
Two-way Switch 2	Second switch, placed at another location such as top of staircase.
Traveller wires	Two wires connecting the two two-way switches.
Lamp	Light controlled from two different positions.
Staircase Wiring	Arrangement that allows ON/OFF control from two places.

23 Special Lighting Connection 3: Fluorescent Lamp Circuit

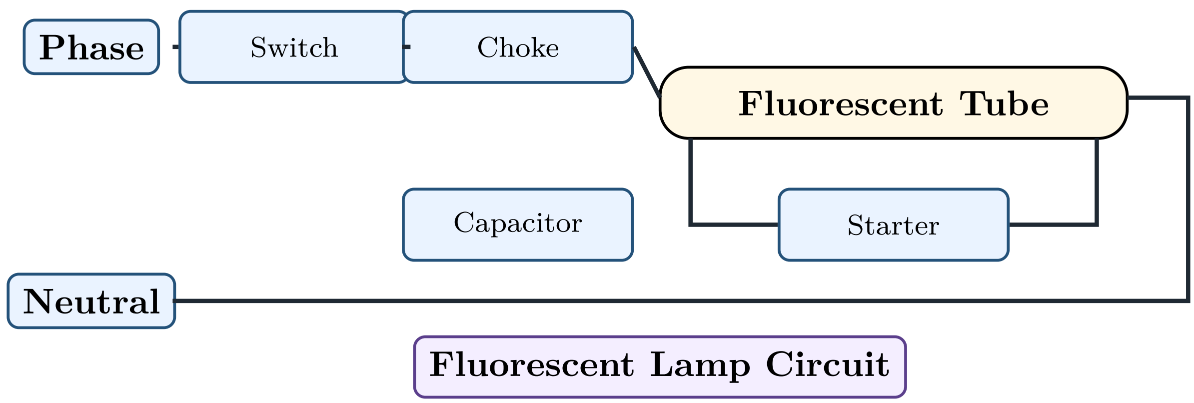
Definition

A **fluorescent lamp circuit** uses a tube light, choke, starter and capacitor to produce light efficiently.

Main parts:

1. Tube light,
2. Choke,
3. Starter,
4. Capacitor,
5. Switch.

24 Fluorescent Lamp Circuit Diagram



How to Read the Diagram

Word/Symbol	Meaning
Phase	Live supply wire.
Neutral	Return supply wire.
Switch	Controls the lamp circuit.
Choke	Inductive coil used to provide high starting voltage and limit current.
Fluorescent Tube	Tube containing gas and fluorescent coating which produces light.
Starter	Helps in starting the tube by preheating electrodes and opening the circuit.
Capacitor	Used for power factor improvement or interference suppression depending circuit.
Fluorescent Lamp Circuit	Complete tube-light connection using choke and starter.

25 Reverse Engineering of Semester Questions

Syllabus Word	Concept	Semester Question Form
Earthing practice	Safety connection to earth	Define earthing and explain its need.
Pipe earthing	GI pipe electrode in earth pit	Draw and explain pipe earthing.
Plate earthing	Copper/GI plate electrode	Draw and explain plate earthing.
Earth resistance	Resistance of electrode and soil	Factors affecting earth resistance.
Service mains	Supplier to consumer connection	Explain domestic service connection.
Wiring system	Methods of building wiring	Compare cleat, batten, casing-capping and conduit wiring.
Sub-circuit wiring	Light and power division	Explain light and power sub-circuits.
Testing installation	Safety tests before energising	List and explain installation tests.
Special lighting	Lamp control circuits	Draw staircase wiring or fluorescent lamp circuit.

26 Most Predictable Semester Questions from Phase 10

No.	Question	Marks
1	Define earthing. Why is earthing necessary?	4–5

2	Draw and explain pipe earthing.	5–6
3	Draw and explain plate earthing.	5–6
4	Compare pipe earthing and plate earthing.	4–5
5	What is earth resistance? State the factors affecting earth resistance.	5
6	State methods to reduce earth resistance.	4
7	Define service mains. Draw domestic service connection diagram.	5–6
8	Explain different types of wiring systems.	6
9	Compare lighting sub-circuit and power sub-circuit.	4
10	What tests are carried out before energising an electrical installation?	5–6
11	Draw one lamp controlled by one switch.	3
12	Draw and explain staircase wiring.	5
13	Draw and explain fluorescent lamp circuit.	5

27 Exam Answer Templates

27.1 Template 1: Earthing Answer

Semester Answer Style

Answer sequence:

1. Define earthing.
2. Write need of earthing.
3. Draw pipe earthing or plate earthing if asked.
4. Explain each part of diagram.
5. Write factors affecting earth resistance.
6. Write methods to reduce earth resistance.

27.2 Template 2: Wiring System Answer

Semester Answer Style

Answer sequence:

1. Define wiring system.
2. Write qualities of good wiring.
3. List types of wiring.
4. Explain each type in one or two lines.
5. Give comparison or use-case table.

27.3 Template 3: Domestic Installation Answer

Semester Answer Style

Answer sequence:

1. Define service mains.
2. Draw domestic connection from distributor to consumer premises.
3. Label energy meter, main switch, DB, light sub-circuit, power sub-circuit and earthing.
4. Explain function of each part.

27.4 Template 4: Testing Answer

Semester Answer Style

Answer sequence:

1. Define testing of installation.
2. Write why testing is necessary.
3. List tests: continuity, insulation resistance, polarity, earth continuity, earth resistance and functional test.
4. Explain each test in one line.

28 Phase 10 Final Memory Sheet

One-Page Memory Sheet

1. Earthing connects metal parts to earth through low resistance path.
2. Earthing protects human life and equipment.
3. Equipment earthing protects metal bodies.
4. System earthing connects neutral or system point to earth.
5. Pipe earthing uses perforated GI pipe.
6. Plate earthing uses copper or GI plate.
7. Charcoal and salt reduce earth resistance.
8. Earth resistance depends on soil, moisture, electrode size and depth.
9. Service mains connect supplier line to consumer premises.
10. Domestic installation has meter, main switch, DB, sub-circuits and earthing.
11. Lighting sub-circuit supplies lamps and fans.
12. Power sub-circuit supplies sockets and heavy loads.
13. Testing includes continuity, insulation, polarity and earth tests.
14. Staircase wiring controls one lamp from two places.
15. Fluorescent lamp circuit uses switch, choke, starter and tube.

29 Phase 10 Self-Test

Answer these without seeing notes:

1. What is earthing?
2. Why is earthing necessary?
3. What is equipment earthing?
4. What is system earthing?
5. Draw pipe earthing.
6. Draw plate earthing.
7. What is earth resistance?
8. What are the factors affecting earth resistance?
9. How can earth resistance be reduced?
10. What are service mains?
11. What are the parts of a domestic electrical installation?
12. What are the common wiring systems?
13. What is a lighting sub-circuit?
14. What is a power sub-circuit?
15. What tests are done before energising an installation?
16. Draw staircase wiring.
17. Draw fluorescent lamp circuit.

Common Mistake

In earthing diagrams, do not just draw the electrode. Always label:

earth electrode	earth wire	charcoal/salt	watering pipe	ground level

These labels carry marks.

End of Phase 10

Syllabus Completion Point

Phase 0 to Phase 10 completes the main Part-II syllabus.

For scoring higher, continue with Phase 11: PYQ Pattern and Question-Family Revision.